Polynomial Coefficient Reference

Complete Numerical Values for All Configurations

December 28, 2025

1 κ Benchmark ($R = 1.3036$)

1.1 PRZZ Baseline

$$\begin{aligned}
P_1:&\quad \text{tilde_coeffs} = [0.261076, -1.071007, -0.236840, 0.260233]\\
P_2:&\quad \text{tilde_coeffs} = [1.048274, 1.319912, -0.940058]\\
P_3:&\quad \text{tilde_coeffs} = [0.522811, -0.686510, -0.049923]\\
Q:&\quad \text{mono_coeffs} = [0.999999, -0.637850, -0.631484, -1.286264, 2.560880, -1.024352]
\end{aligned}$$

Result: $\kappa = 0.41729593$, $c = 2.13744891$

1.2 Optimized ($\alpha = 70$, $\beta = -30$)

$$\begin{aligned}
P_1:&\quad \text{tilde_coeffs} = [0.261076, -1.071007, -0.236840, 0.260233]\\
P_2:&\quad \text{tilde_coeffs} = [1.182524, -0.203070, -0.217819]\\
P_3:&\quad \text{tilde_coeffs} = [-1.085241, -2.260095, -0.125575]\\
Q:&\quad \text{mono_coeffs} = [0.999999, -0.637850, -0.631484, -1.286264, 2.560880, -1.024352]
\end{aligned}$$

Result: $\kappa = 0.46196550$, $c = 2.01653719$
Improvement: $+10.70\%$

2 κ^* Benchmark ($R = 1.1167$)

2.1 PRZZ Baseline

$$\begin{aligned}
P_1:&\quad \text{tilde_coeffs} = [0.052703, -0.657999, -0.003193, -0.101832]\\
P_2:&\quad \text{tilde_coeffs} = [1.049837, -0.097446]\\
P_3:&\quad \text{tilde_coeffs} = [0.035113, -0.156465]\\
Q:&\quad \text{mono_coeffs} = [1.000000, -1.032446, 0, 0, 0, 0]
\end{aligned}$$

Result: $\kappa^* = 0.40750979$, $c = 1.93795602$

2.2 Optimized ($\alpha = 2.75$)

$$\begin{aligned}
P_1 :&\quad \text{tilde_coeffs} = [0.052703, -0.657999, -0.003193, -0.101832] \\
P_2 :&\quad \text{tilde_coeffs} = [1.344449, -0.063203, 0.000000] \\
P_3 :&\quad \text{tilde_coeffs} = [0.032372, -0.146727, 0.000000] \\
Q :&\quad \text{mono_coeffs} = [1.000000, -1.032446, 0.000000, 0.000000, 0.000000, 0.000000]
\end{aligned}$$

Result: $\kappa^* = 0.42761669$, $c = 1.89492724$
Improvement: $+4.93\%$

3 Polynomial Basis Conventions

3.1 P_1 Polynomial

P_1 uses a constrained form satisfying $P_1(0) = 0$, $P_1(1) = 1$:

$$P_1(x) = x + x(1 - x) \cdot \tilde{P}_1(1 - x) \tag{1}$$

where $\tilde{P}_1$ is in powers of $(1 - x)$:

$$\tilde{P}_1(y) = c_0 + c_1 y + c_2 y^2 + c_3 y^3 \tag{2}$$

3.2 P_ℓ Polynomials ($\ell \geq 2$)

P_ℓ satisfies $P_\ell(0) = 0$:

$$P_\ell(x) = x \cdot \tilde{P}_\ell(x) \tag{3}$$

where $\tilde{P}_\ell$ is in monomial basis:

$$\tilde{P}_\ell(x) = c_0 + c_1 x + c_2 x^2 + \cdots \tag{4}$$

3.3 Q Polynomial

Q is stored in monomial basis:

$$Q(x) = q_0 + q_1 x + q_2 x^2 + \cdots + q_5 x^5 \tag{5}$$

PRZZ also uses a $(1 - 2x)^k$ basis representation with constraint $Q(0) = 1$.